

Grid Features Defense Packet -- XB2IQX

Every trace in this packet is SIMULATED output from a fitted conductance-based single-compartment cell model (src/singlecell_model_v1.py), integrated deterministically at a fixed timestep -- not a raw current-clamp recording. What's being defended here is the feature-extraction methodology applied on top of that model (spike detection, burst/tonic classification, rebound detection, sag depth, adaptation ratio) -- i.e. do these algorithms correctly characterize the dynamics the model actually produces. Every panel below reruns the exact production function on a real simulated trace and marks what it found, so the classification isn't just a text label to take on faith.

cell_floor = -4.50 nA | n grid points = 2500 | burstiness index = 0.042 | F-I slope = 19.91 Hz/nA ($R^2=0.976$)

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1. Spike detection

Every detected peak overlaid on real traces.

2. Prominence-threshold sensitivity

How stable spike counts are across a range of thresholds.

3. dV/dt cross-validation

Independent detector cross-check across a random sample.

4. Tonic/bursting/silent classification

ISI sequence + log-ISI KDE evidence.

5. Rebound detection

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6. Sag depth

Baseline, search window, and trough on a real trace.

7. Adaptation ratio

First/last ISI windows on a real, sustained tonic trace.

8. Firing rate / F-I slope

Rate vs. injected current with the fit line.

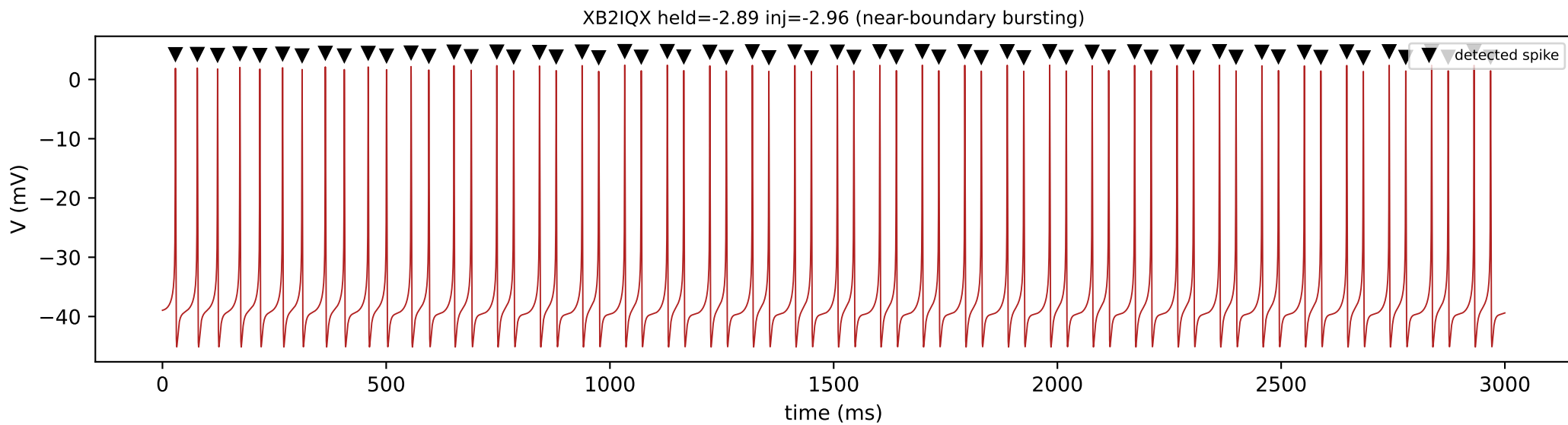
9. Self-consistency check

Resimulated vs. cached classification match rate.

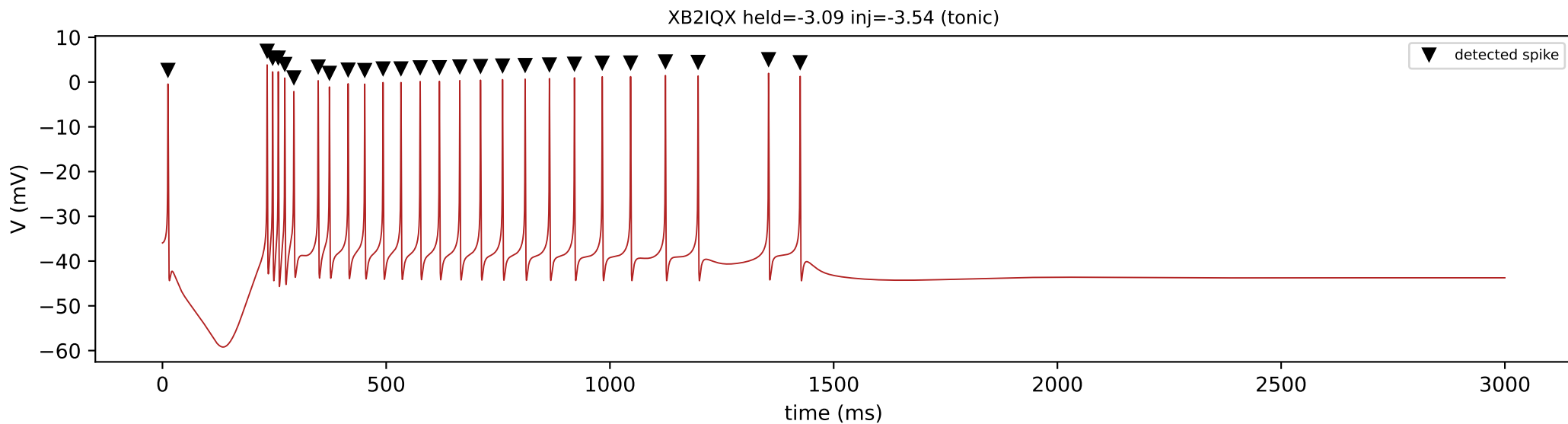
10. Burst-onset-then-silence detection

Onset-burst detection and the adaptation-ratio ceased-firing gate, contrasted against a burst-onset that keeps firing.

1. Spike detection

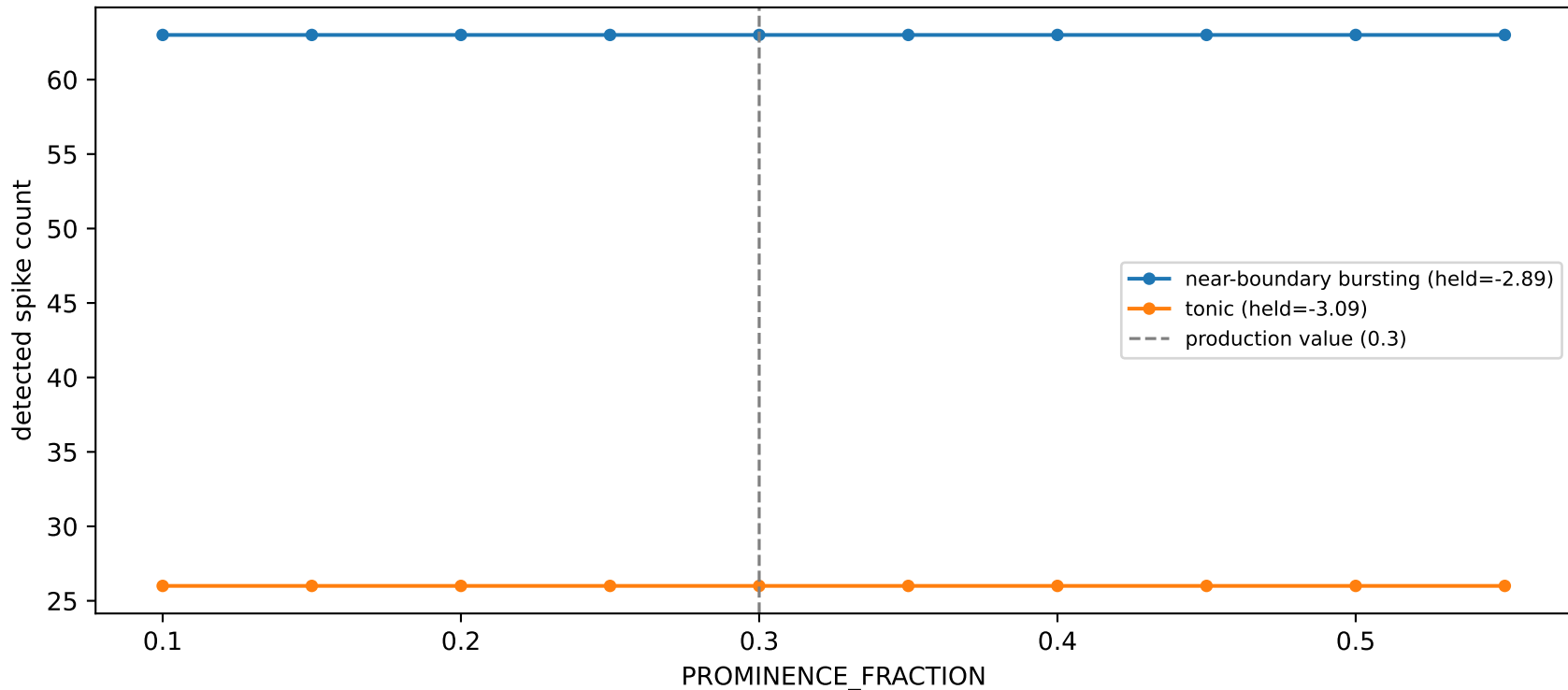


Markers show every peak `scipy.signal.find_peaks` detects at prominence = 0.30 x this trace's own voltage range (47.6 mV) -- the exact call `count_spikes_and_rate/compute_isi_ms` use for every spike count, ISI, and burst/tonic classification in the pipeline. 63 spike(s) detected in this window.



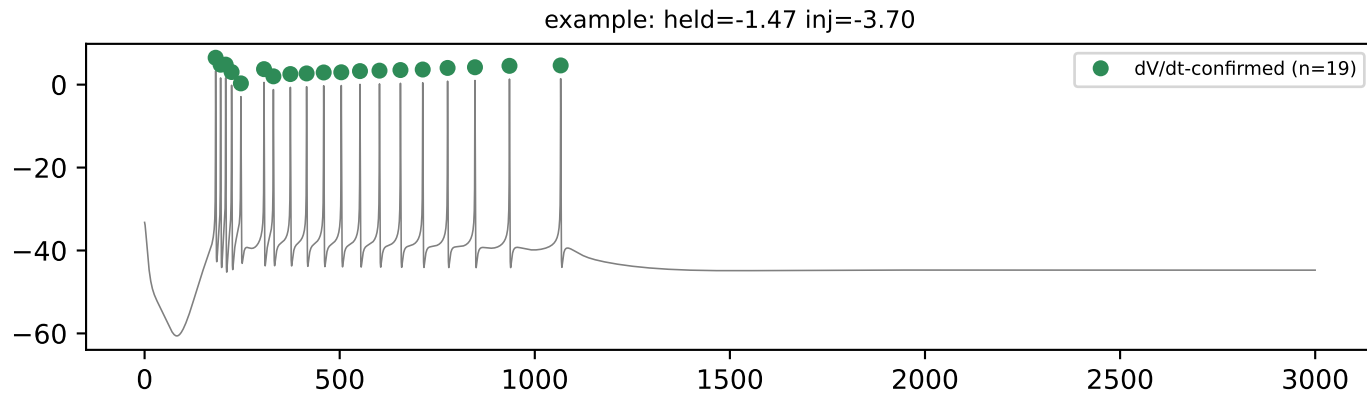
2. Prominence-threshold sensitivity

XB2IQX -- spike count vs. prominence threshold



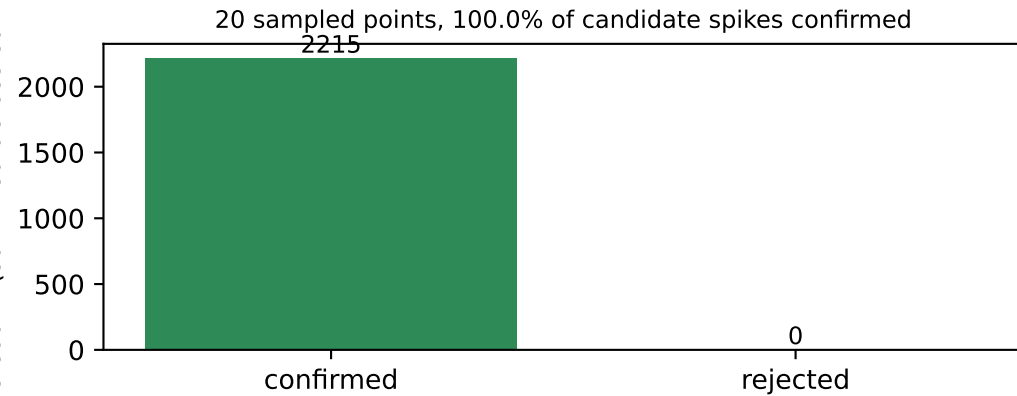
PROMINENCE_FRACTION=0.3 is the single most load-bearing, least-independently-calibrated constant in the pipeline -- every spike count, ISI, and burst/tonic call across the codebase traces back to one `find_peaks(..., prominence=range*0.3)` call. This sweeps that threshold from 0.10 to 0.55 on real traces: a flat plateau around the production value means the spike count this pipeline reports is not sensitive to the exact threshold chosen, not an arbitrary cliff-edge pick.

3. dV/dt cross-validation (unused-in-production detector as a sanity check)



Cross-check against a second, unused-in-production detector: a candidate peak additionally requires a genuine fast upstroke ($dV/dt \geq 10$ mV/ms within 20 ms before the peak, Bean 2007 convention) before counting. 19 of 19 candidate peak(s) confirmed; the two detectors agree on every spike in this window.

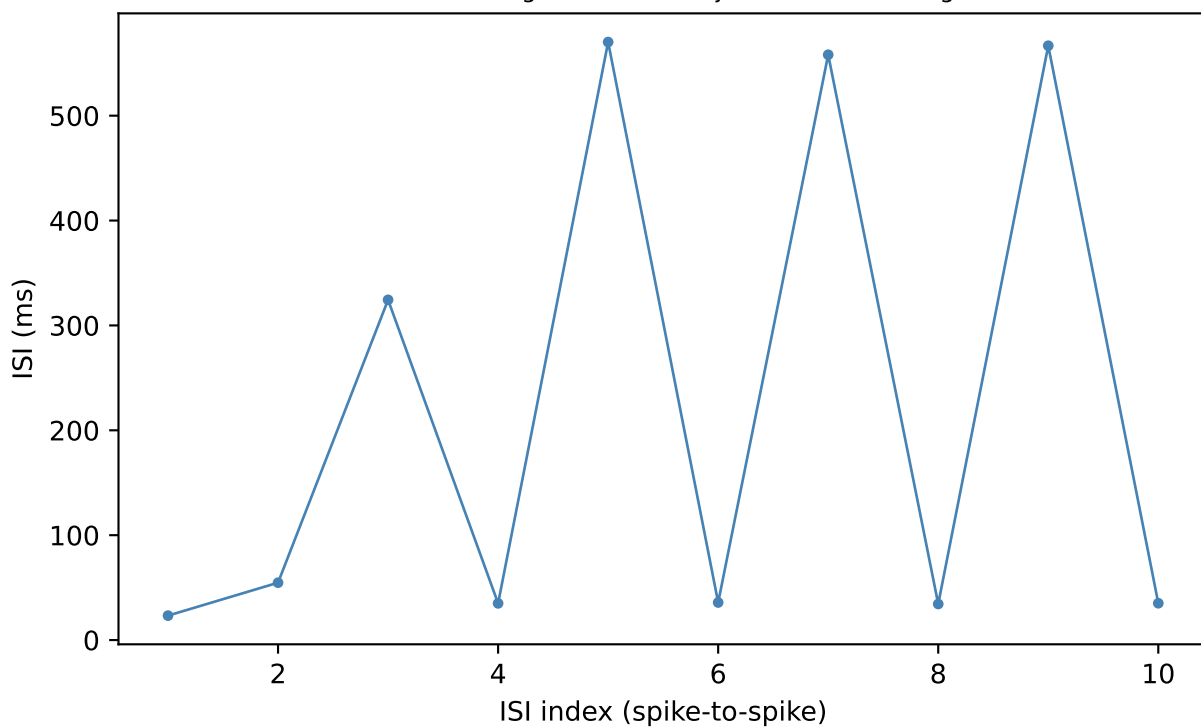
spike count (summed across sample)



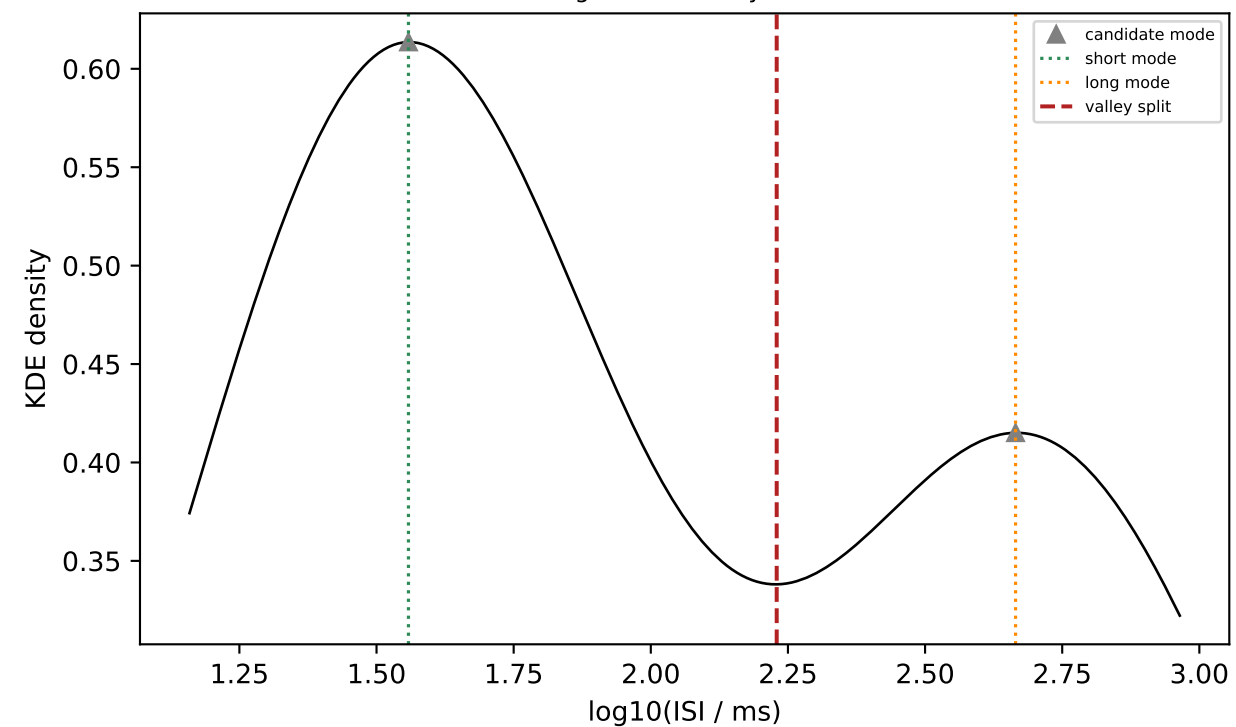
Cross-check across a random sample of 20 XB2IQX grid points: every candidate prominence-based peak is additionally required to pass a dV/dt shape-confirmation gate (`detect_spikes_dvdt_confirmed` -- present in the codebase but not called by the production pipeline). Agreement close to 100% means the simpler production detector isn't missing a shape check that changes the answer.

4. Tonic / bursting / silent classification

clean bursting: held=-3.21 inj=-3.36 -> 'bursting'

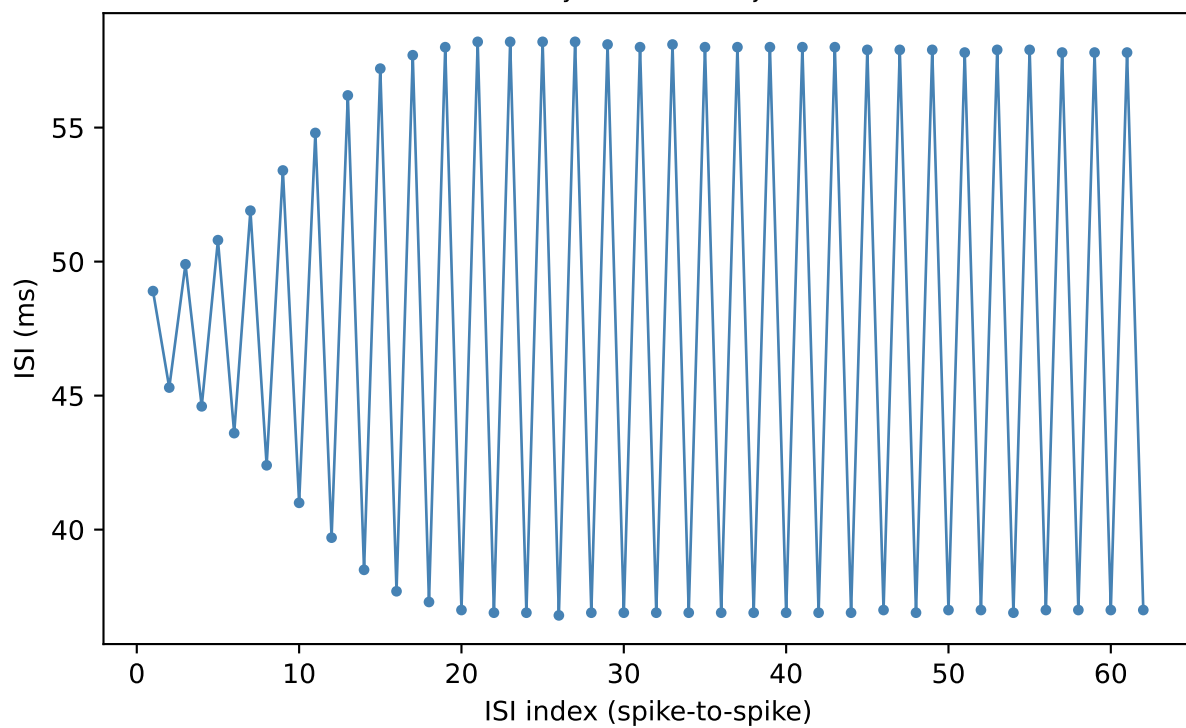


log-ISI bimodality test

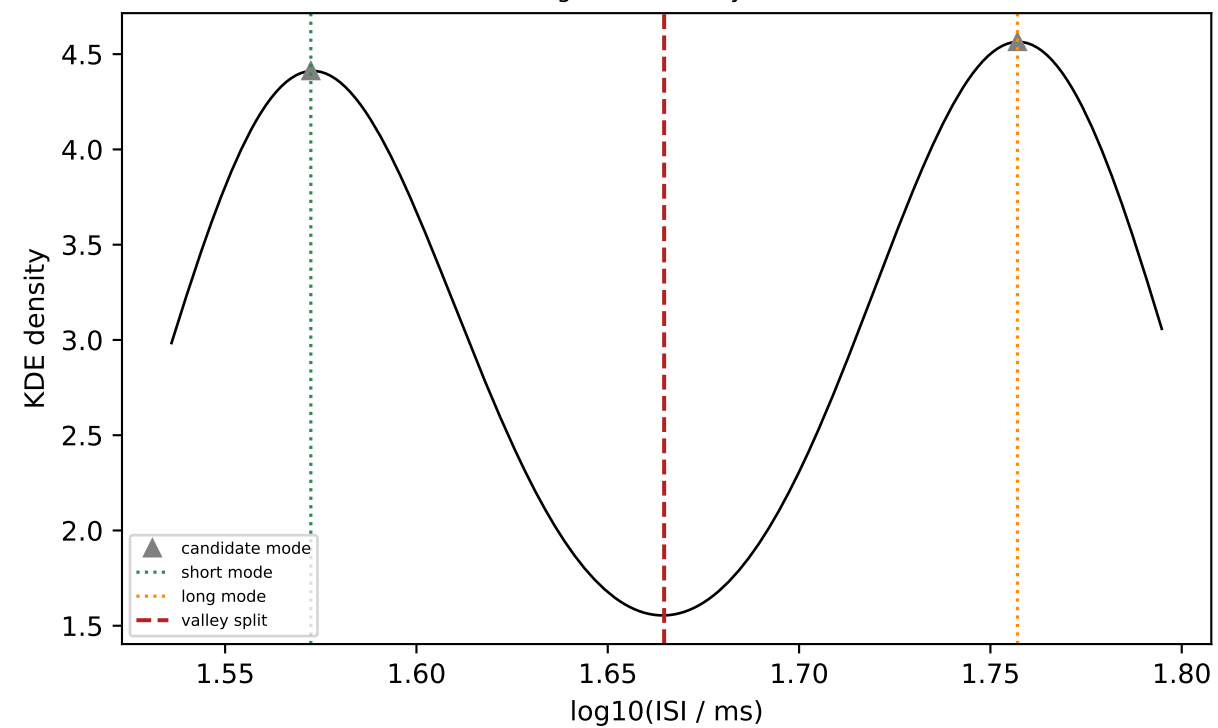


11 spikes, 10 ISIs. Split at 2 candidate mode(s) -> short
ISI=36.4 ms, long ISI=504.8 ms (ratio 13.88x, needs >= 1.5x). Avg spikes/burst = 2.20 (needs >= 1.5). Final call:
'bursting' (bimodality metric / Ashman's D = 6.33).

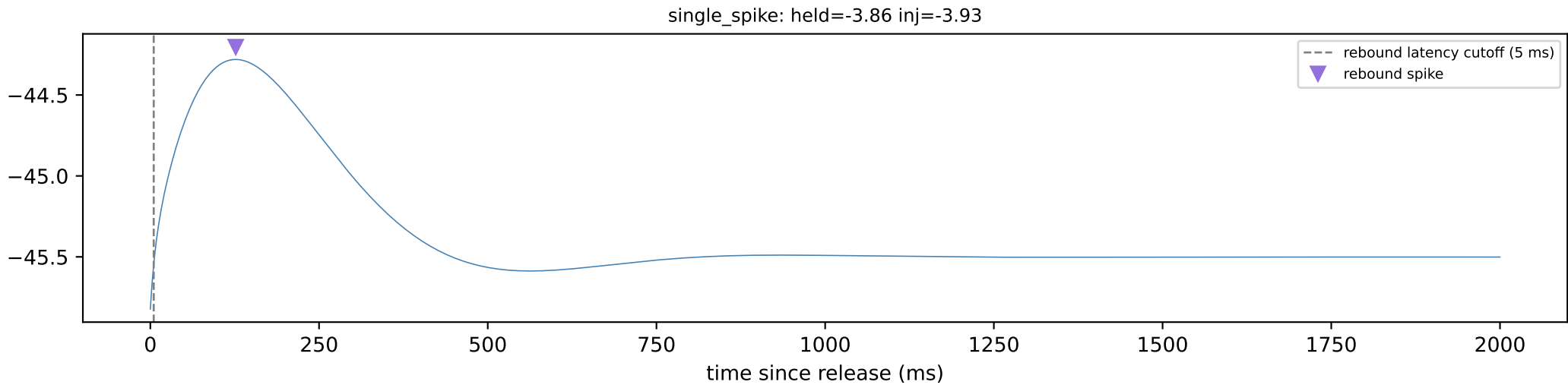
near-miss boundary: held=-2.89 inj=-2.96 -> 'tonic'



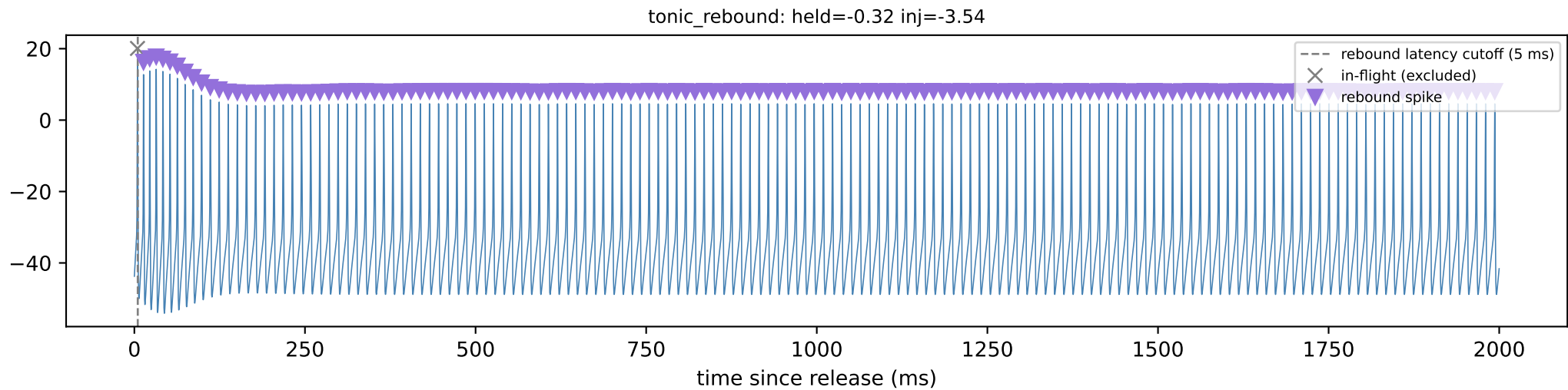
log-ISI bimodality test



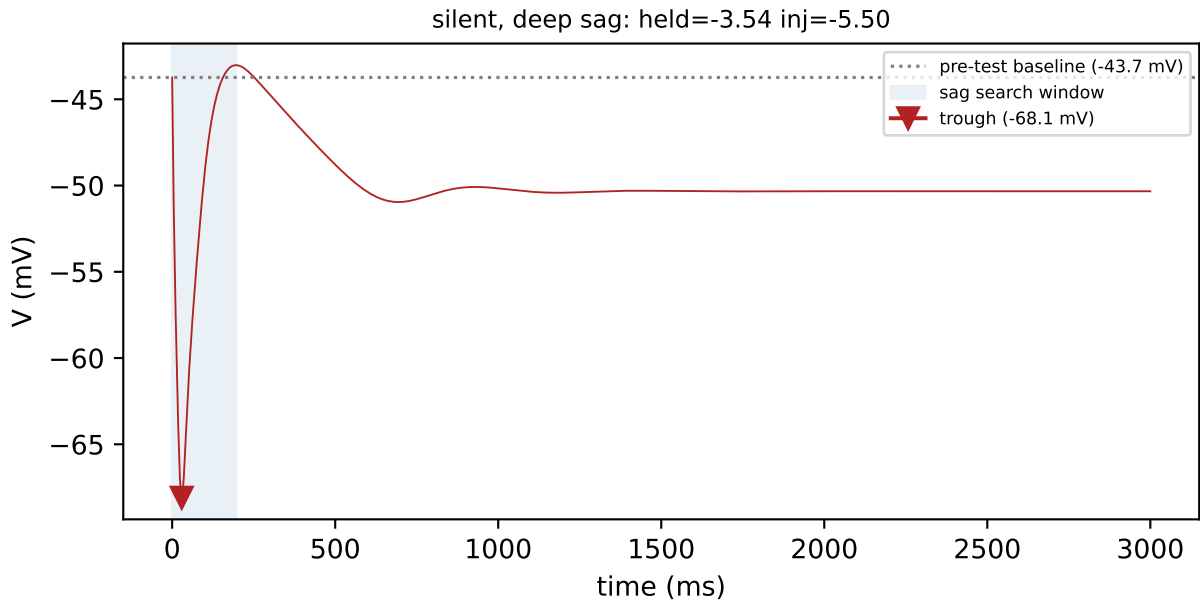
5. Rebound (post-inhibitory) detection



1 spike(s) at or after the 5 ms cutoff count as rebound. rebound_occurred=True.

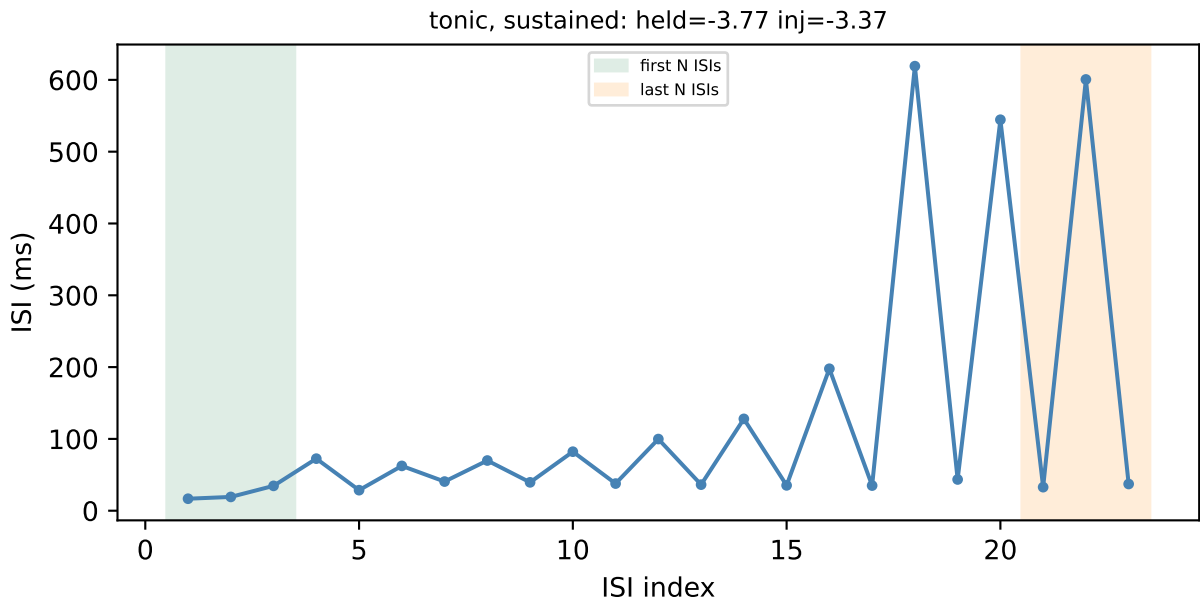


6. Sag depth



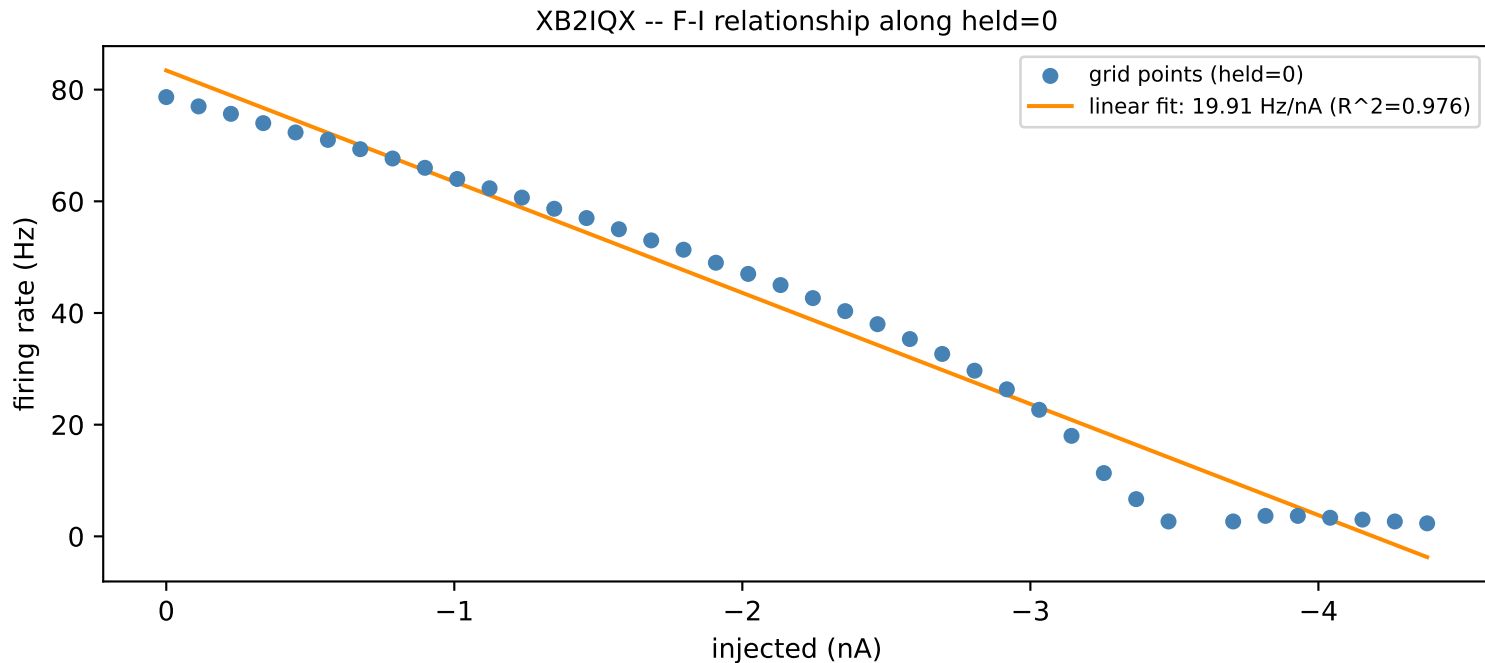
Sag depth = baseline - trough = $-43.7 - (-68.1) = 24.4$ mV. Baseline is the held-current settle's final voltage; trough is the minimum voltage over truncated early by the first spike at 196 ms.

7. Adaptation ratio



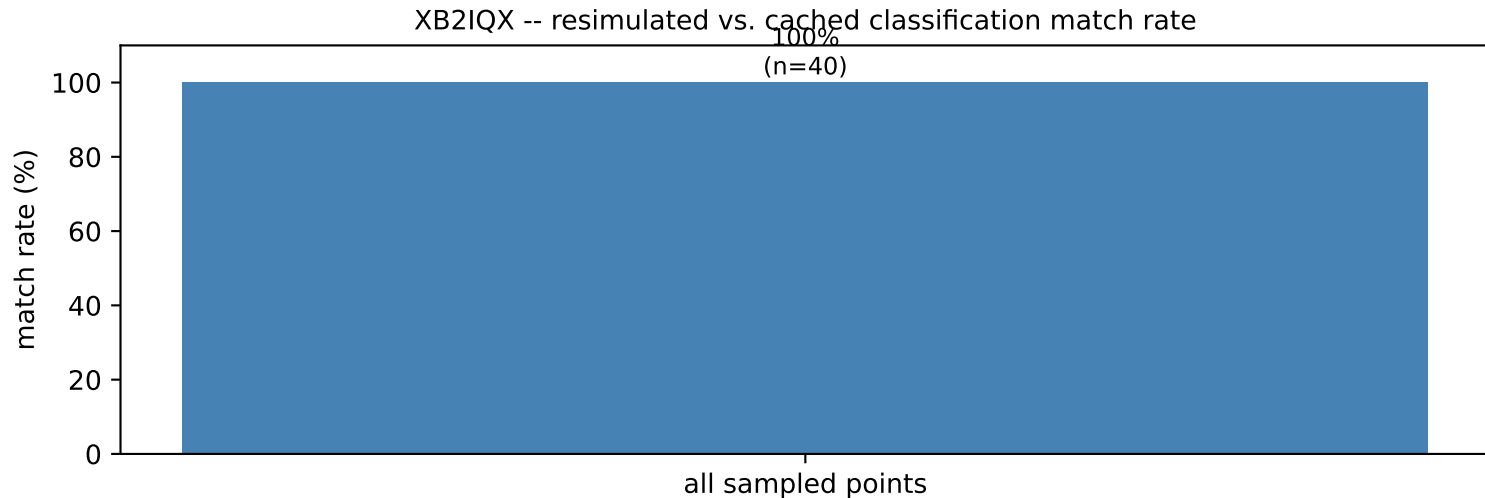
Adaptation ratio = $\text{mean}(\text{last 3 ISIs}) / \text{mean}(\text{first 3 ISIs}) = 9.51$. Firing slows (adapts) over the course of this test window.

8. Firing rate / F-I slope



39 non-silent, confidently-classified points along the held=0 column (compute_fi_slope's own filter -- flat zero-rate points below rheobase-equivalent excluded so they don't bias the linear fit). This scalar (fi_slope_hz_per_nA) already feeds the cross-cell feature table; this is simply the first time it's been plotted against the data it was fit to.

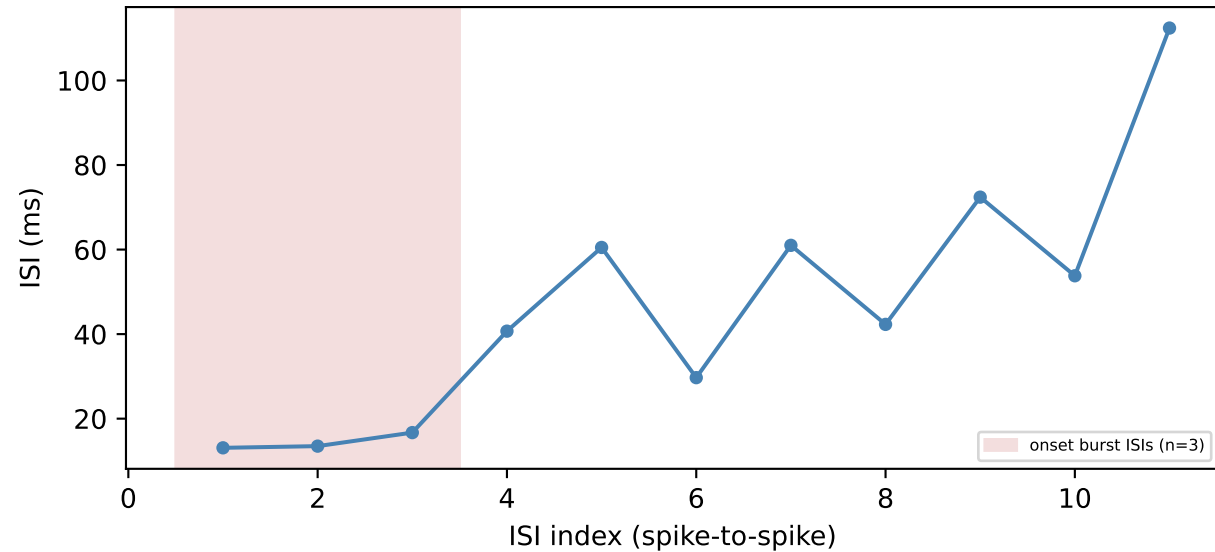
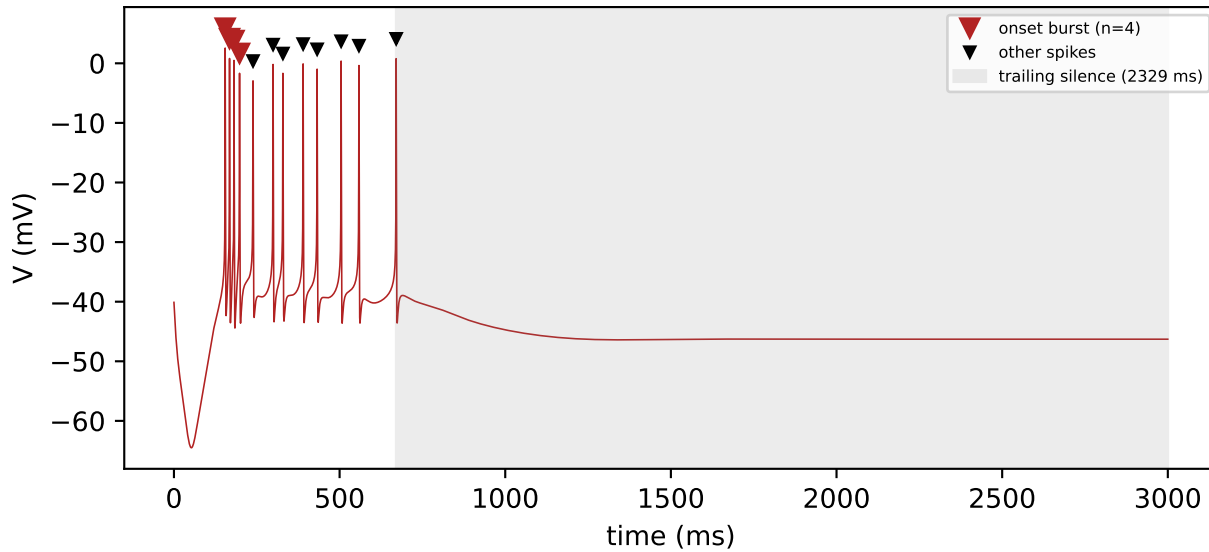
9. Self-consistency check



Re-simulates 40 random cached grid points end-to-end and checks whether the resulting test_pattern/rebound_pattern still matches what's stored in cell_held_injected_grid.pkl. Match rate: 100%. Every held level in the current uniform-grid sweep (except held=0, which reuses the cell's own cached limit-cycle state exactly) is settled via a warm-start from whichever already-settled held level is numerically nearest -- the module docstring in run_held_injected_grid.py already documents that this model shows path-dependent (hysteretic) settling near dynamical transitions, so a resimulation that warm-starts from a different already-settled state than the original sweep used can land on a different classification near a boundary. Any mismatches here are consistent with that known property, not a new bug.

10. Burst-onset-then-silence detection

burst then ceased: held=-2.48 inj=-4.04 -> pattern='tonic', adaptation_ratio=None (suppressed)



Onset burst detected: 4 spikes (mean ISI 14.4 ms) at window start, found by scanning locally for the first ISI jump -- independent of whatever whole-window tonic/bursting/silent label the KDE bimodality test assigns. Trailing silence after the last spike: 2329 ms of a 3000 ms window. likely_ceased_firing=True -- 2329 ms is $\geq 3 \times$ the most recent ISI, so test_adaptation_ratio is suppressed (reported as None): a first-k/last-k ratio isn't a meaningful 'smooth adaptation' number for a train that stopped partway through the window.

burst then sustained: held=-3.77 inj=-3.37 -> pattern='tonic', adaptation_ratio=9.51

